

1 考点 1: 信用风险管理

1.

A.

答案:B

解析:

A 选项错误。

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B 选项正确。

•

C 选项错误。

•

D 选项错误。

•

考察:

2 考点 2: 衍生品信用风险

1.

A.

答案:B

解析:

A 选项错误。

•

B 选项错误。

•

C 选项错误。

•

D 选项错误。

•

考察:

3 考点 3: 结构化信用产品

1. Which of the following statements regarding frictions in the securitization of subprime mortgages is correct?
- A. The arranger will typically have an information advantage over the originator with regard to the quality of the loans securitized
- B. The originator will typically have an information advantage over the arranger, which can create an incentive for the originator to collaborate with the borrower in filing false loan applications
- C. The major credit rating agencies are paid by investors for their rating service of mortgage-backed securities, and this creates a potential conflict of interest
- D. The use of escrow accounts for insurance and tax payments eliminates the risk of foreclosure

答案:B

解析:

A 选项错误。这个说法不正确:

- 安排人通常不会比发起人更了解贷款质量
- 发起人作为贷款的直接发放者具有信息优势
- 这个选项误解了信息不对称的方向

B 选项正确。这个说法正确:

- 发起人确实比安排人更了解贷款质量
- 这种信息优势可能导致发起人与借款人合谋
- 发起人可能协助借款人伪造贷款申请材料
- 这种欺诈行为会增加证券化产品的风险

C 选项错误。这个说法不正确:

- 评级机构不是由投资者支付评级费用
- 评级费用通常由证券化产品的发行方支付
- 这个选项误解了评级机构的收费模式

D 选项错误。这个说法不正确:

- 托管账户只能延缓但不能消除止赎风险
- 这个选项夸大了托管账户的作用
- 这个选项不符合实际情况

考察: 次贷证券化中发起人具有信息优势, 可能引发欺诈风险

2. Assume there are 100 identical loans with a principal balance of \$500,000 each. Based on a credit analysis, a 300 basis point spread is applied to the borrowers. LIBOR is currently 4% and the coupon rate will reset annually. The senior, junior, and equity tranches are 75%, 20%, and 5% of the pool, respectively. The spreads on the senior and mezzanine tranches are 2% and 6%. Excess cash flow is diverted to trust account above \$1,000,000. Assume the default rate is 2%. What are the cash flows to the mezzanine and trust account in the first period?

- A. \$1,000,000 \$0
- B. \$1,000,000 \$180,000
- C. \$2,250,000 \$200,000

D. \$2,250,000 \$250,000

答案:A

解析:

A 选项正确。这个说法正确:

- $\text{贷款利率} = 4\%(\text{LIBOR}) + 3\%(\text{利差}) = 7\%$
- $\text{第一期总现金流} = 100 \times \$500,000 \times 7\% \times (1 - 2\%) = \$3,430,000$
- $\text{高级层现金流} = \$50\text{M} \times 75\% \times (4\% + 2\%) = \$2,250,000$
- $\text{夹层级现金流} = \$50\text{M} \times 20\% \times (4\% + 6\%) = \$1,000,000$
- $\text{剩余现金流} = \$3,430,000 - \$2,250,000 - \$1,000,000 = \$180,000$
- 由于剩余现金流小于 \$1,000,000，全部归股权层，信托账户为 0

B 选项错误。这个说法不正确:

- 夹层级现金流计算正确
- 但剩余现金流应归股权层，不应进入信托账户
- 这个选项误解了现金流分配规则

C 选项错误。这个说法不正确:

- 高级层现金流计算正确
- 但夹层级现金流计算错误
- 这个选项完全误解了现金流分配

D 选项错误。这个说法不正确:

- 高级层现金流计算正确
- 但夹层级现金流和信托账户金额都计算错误
- 这个选项完全误解了现金流分配

考察: 证券化产品现金流分配遵循优先级顺序，剩余现金流归股权层

3. Assume a firm issues only three capital claims: zero-coupon senior debt with face value of \$300 million; zero-coupon junior debt with face value of \$500 million; and \$200 million in equity. What is the credit enhancement provided to the senior debt?

- A. \$200 million
- B. \$300 million
- C. \$500 million
- D. \$700 million

答案:D

解析:

A 选项错误。这个说法不正确:

- 仅考虑了股权价值作为信用增级
- 忽略了次级债的信用增级作用
- 这个选项低估了信用增级总额

B 选项错误。这个说法不正确:

- 仅考虑了高级债的面值
- 这个选项完全误解了信用增级的概念
- 这个选项与问题无关

C 选项错误。这个说法不正确：

- 仅考虑了次级债的面值
- 忽略了股权的信用增级作用
- 这个选项低估了信用增级总额

D 选项正确。这个说法正确：

- 信用增级来自次级债和股权的价值
- 次级债面值 \$500 百万
- 股权价值 \$200 百万
- 总信用增级 = \$500 百万 + \$200 百万 = \$700 百万
- 这些资金在高级债之前承担损失

考察：高级债的信用增级来自次级债和股权的价值总和

4. A pool of high yield bonds is placed in an SPV and three tranches (including the equity tranche) of bonds are issued collateralized by the bonds to create a Collateralized Bond Obligation (CBO). Which of the following is true?

- A. At fair value, the value of the issued bonds should be less than the collateral
- B. At fair value, the total default probability, weighted by size of issue, of the issued bonds should equal the default probability of the collateral pool
- C. The equity tranche of the CBO has the least risk of default
- D. The yield on the low risk tranche must be greater than the yield on the collateral pool

答案:B

解析：

A 选项错误。这个说法不正确：

- 在公允价值下，发行的债券价值应等于抵押品价值
- 这个选项误解了 CBO 的基本定价原则
- 这个选项不符合无套利原则

B 选项正确。这个说法正确：

- CBO 和基础证券必须具有相同的市场价值
- 加权后的违约概率必须相等
- 现金流模式必须相似
- 风险特征必须一致

C 选项错误。这个说法不正确：

- 股权层级具有最高的违约风险
- 这个选项完全误解了 CBO 的风险分配
- 这个选项与实际情况相反

D 选项错误。这个说法不正确：

- 低风险层级的收益率必须低于抵押品池的收益率
- 这个选项误解了 CBO 的收益分配
- 这个选项与实际情况相反

考察：CBO 与基础证券必须具有相同的市场价值和风险特征

5. Continuously increasing default probability (while holding default correlation constant) will most likely have what effect on the credit VaR of mezzanine and equity tranches?

- A. Increase Increase then decrease
- B. Increase Decrease then increase
- C. Decrease Increase then decrease
- D. Decrease Decrease then increase

答案:C

解析：

A 选项错误。这个说法不正确：

- 股权层 VaR 会随着违约概率上升而下降
- 夹层级 VaR 的变化趋势描述错误
- 这个选项完全误解了 VaR 的变化特征

B 选项错误。这个说法不正确：

- 股权层 VaR 会随着违约概率上升而下降
- 夹层级 VaR 的变化趋势描述错误
- 这个选项完全误解了 VaR 的变化特征

C 选项正确。这个说法正确：

- 股权层 VaR 下降原因：
 - 违约概率上升时，股权层损失确定性增加
 - 损失波动性减小，导致 VaR 下降
- 夹层级 VaR 先升后降原因：
 - 低违约概率时，表现类似高级层，VaR 上升
 - 高违约概率时，表现类似股权层，VaR 下降

D 选项错误。这个说法不正确：

- 股权层 VaR 变化趋势正确
- 但夹层级 VaR 的变化趋势描述错误
- 这个选项部分正确但整体错误

考察：违约概率上升时，股权层 VaR 下降，夹层级 VaR 先升后降

6. Which of the following statements on credit enhancement on securitized assets for the purposes of mitigating credit risk and liquidation risk are true? I. Internal and external enhancements can be used for both credit and liquidity risk. II. Liquidity reserves for liquidity risk are similar to a cash collateral account for credit risk.

- A. I only
- B. II only
- C. Both I and II
- D. Neither I nor II

答案:C

解析:

A 选项错误。这个说法不正确:

- 仅考虑了内部和外部增级措施
- 忽略了流动性储备与现金抵押账户的相似性
- 这个选项不完整

B 选项错误。这个说法不正确:

- 仅考虑了流动性储备与现金抵押账户的相似性
- 忽略了内部和外部增级措施
- 这个选项不完整

C 选项正确。这个说法正确:

- 陈述 I 正确:
 - 信用风险和流动性风险都可以使用内部增级
 - 信用风险和流动性风险都可以使用外部增级
- 陈述 II 正确:
 - 流动性储备用于应对流动性风险
 - 现金抵押账户用于应对信用风险
 - 两者在功能上具有相似性

D 选项错误。这个说法不正确:

- 两个陈述都是正确的
- 这个选项完全错误
- 这个选项误解了增级措施的特征

考察: 证券化产品可通过内部外部增级措施应对信用和流动性风险

7. An important step in structuring a securitization is determining an adequate level of credit support or enhancement. Which of the following is not one of common credit enhancement forms for securitization transactions?

- A. Subordinated tranches of securitization debt
- B. Excess Spread account
- C. Cash reserve account
- D. Sinking fund account

答案:D

解析:

A 选项错误。这个说法不正确:

- 次级层级是常见的信用增级形式
- 通过优先级顺序提供信用保护
- 这个选项是有效的信用增级方式

B 选项错误。这个说法不正确：

- 超额利差账户是常见的信用增级形式
- 通过资产池收益与支出差额提供保护
- 这个选项是有效的信用增级方式

C 选项错误。这个说法不正确：

- 现金储备账户是常见的信用增级形式
- 通过预留现金提供信用保护
- 这个选项是有效的信用增级方式

D 选项正确。这个说法正确：

- 偿债基金不是信用增级形式
- 偿债基金用于定期赎回债券
- 偿债基金是债券发行的常规安排
- 偿债基金与信用增级无关

考察：偿债基金不是证券化产品的信用增级形式

8. Which of the following is not a friction in the subprime securitization market?

- A. Investor and rating agency
- B. Servicer and mortgagor
- C. Mortgagor and arranger
- D. Asset manager and investor

答案:C

解析：

A 选项错误。这个说法不正确：

- 投资者与评级机构之间存在摩擦
- 评级机构可能低估风险
- 投资者可能过度依赖评级
- 这个选项是次贷证券化市场的摩擦之一

B 选项错误。这个说法不正确：

- 服务商与借款人之间存在摩擦
- 服务商可能缺乏激励去管理贷款
- 借款人可能面临服务问题
- 这个选项是次贷证券化市场的摩擦之一

C 选项正确。这个说法正确：

- 借款人与承销人没有直接接触
- 两者之间不存在直接关系
- 因此不存在摩擦
- 这个选项不是次贷证券化市场的摩擦

D 选项错误。这个说法不正确：

- 资产管理人与投资者之间存在摩擦
- 可能存在利益冲突
- 可能存在信息不对称
- 这个选项是次贷证券化市场的摩擦之一

考察：次贷证券化市场中借款人与承销人无直接接触，不存在摩擦

9. A hedge fund is considering taking positions in various tranches of a collateralized debt obligation (CDO). The fund's chief economist predicts that the default probability will decrease significantly and that the default correlation will increase. Based on this prediction, which of the following is a good strategy to pursue?

- A. Buy the senior tranche and buy the equity tranche
- B. Buy the senior tranche and sell the equity tranche
- C. Sell the senior tranche and sell equity tranche
- D. Sell the senior tranche and buy the equity tranche

答案:D

解析：

A 选项错误。这个说法不正确：

- 违约概率下降对股权层有利
- 但违约相关性上升对高级层不利
- 这个策略没有考虑违约相关性的影响

B 选项错误。这个说法不正确：

- 违约概率下降对股权层有利
- 不应该做空股权层
- 这个策略完全误解了市场预期

C 选项错误。这个说法不正确：

- 违约概率下降对股权层有利
- 不应该做空股权层
- 这个策略完全误解了市场预期

D 选项正确。这个说法正确：

- 违约概率下降的影响：
 - 提高股权层价值
 - 降低违约风险
- 违约相关性上升的影响：

- 提高股权层价值
- 降低高级层价值
- 因此应该做多股权层，做空高级层

考察: 违约概率下降和相关性上升时，应做多股权层做空高级层

10. Which of the following subprime characteristics provide direct protection for senior tranches?

- A. Subordination, excess spread, and shifting interest
- B. Subordination, prepayments, and shifting interest
- C. Overcollateralization, excess spread, and tinning of losses
- D. Overcollateralization, excess spread, and prepayments

答案:A

解析:

A 选项正确。这个说法正确:

- 分层 (Subordination):
 - 通过优先级顺序提供保护
 - 次级层级先承担损失
- 超额利差 (Excess Spread):
 - 资产池收益大于支出
 - 提供额外的信用保护
- 利率锁定期 (Shifting Interest):
 - 锁定期内本金优先偿付高级层
 - 中间层只能获得利息

B 选项错误。这个说法不正确:

- 提前还款不是直接保护机制
- 可能加速或延缓现金流
- 这个选项误解了保护机制

C 选项错误。这个说法不正确:

- 损失时间不是直接保护机制
- 只影响超额利差
- 这个选项误解了保护机制

D 选项错误。这个说法不正确:

- 提前还款不是直接保护机制
- 可能加速或延缓现金流
- 这个选项误解了保护机制

考察: 次贷证券化中分层、超额利差和利率锁定期为高级层提供直接保护

11. Which of the following statements most accurately describes the effect of selling a loan without recourse?

- A. The bank that sells the loan retains a contingent liability
- B. The bank that sells the loan bears a specified percentage of the credit risk
- C. The loan is removed from the balance sheet of the bank that sells the loan
- D. The purchaser has the right to sell the loan back to the bank that originated the loan

答案:C

解析:

A 选项错误。这个说法不正确:

- 无追索权出售意味着银行不保留或有负债
- 所有信用风险都转移给买方
- 这个选项误解了无追索权的含义

B 选项错误。这个说法不正确:

- 无追索权出售意味着银行不承担任何信用风险
- 所有信用风险都转移给买方
- 这个选项误解了无追索权的含义

C 选项正确。这个说法正确:

- 无追索权出售的影响:
 - 贷款从银行资产负债表中移除
 - 所有信用风险转移给买方
 - 银行不再承担任何风险

D 选项错误。这个说法不正确:

- 无追索权出售意味着买方不能将贷款卖回给银行
- 这个选项误解了无追索权的含义
- 这个选项与实际情况相反

考察: 无追索权出售贷款时, 贷款从银行资产负债表中移除

12. During the recent credit crisis, subprime mortgages received pressure from upward movements in interest rates, along with the impossibility for most of these borrowers to refinance. As a result, many subprime borrowers allowed their reduced value homes to be taken over by the lender. All of the following statements describe the effect subprime actions had on mortgage-backed security portfolios except:

- A. Repayment issues in lower tranches impacted confidence in higher tranches
- B. There was a fire sale of securitized debt
- C. Panic among MBS investors led to a flight to safer assets
- D. The defaults in the lower tranches were high but did not affect senior tranche investors

答案:D

解析:

A 选项错误。这个说法不正确:

- 次级层级还款问题确实影响了高级层信心
- 这是次贷危机的典型特征

- 这个选项描述了实际发生的情况

B 选项错误。这个说法不正确：

- 确实发生了证券化债务的甩卖
- 这是市场恐慌的表现
- 这个选项描述了实际发生的情况

C 选项错误。这个说法不正确：

- MBS 投资者确实出现恐慌
- 确实转向更安全的资产
- 这个选项描述了实际发生的情况

D 选项正确。这个说法正确：

- 这个说法不符合实际情况：
 - 次级层级违约确实影响了高级层投资者
 - 信心危机蔓延到所有层级
 - 高级层投资者也受到损失

考察：次贷危机中次级层级违约影响了所有层级的投资者

13. Which of the following statements about portfolio losses and default correlation are most likely correct? I. Increasing default correlation decreases senior tranche values but increases equity tranche values. II. At high default rates, increasing default correlation decreases mezzanine bond prices.

- A. I only
- B. II only
- C. Both I and II
- D. Neither I nor II

答案:A

解析：

A 选项正确。这个说法正确：

- 陈述 I 正确：
 - 违约相关性上升增加极端组合收益的可能性
 - 高违约可能性降低高级层价值
 - 低违约可能性提高股权层价值
- 陈述 II 错误：
 - 高违约率时，相关性上升增加极端收益可能性
 - 这对股权层和夹层级都有利
 - 不会降低夹层级债券价格

B 选项错误。这个说法不正确：

- 陈述 I 是正确的
- 陈述 II 是错误的
- 这个选项部分正确但整体错误

C 选项错误。这个说法不正确：

- 陈述 I 是正确的
- 陈述 II 是错误的
- 这个选项部分正确但整体错误

D 选项错误。这个说法不正确：

- 陈述 I 是正确的
- 陈述 II 是错误的
- 这个选项完全错误

考察：违约相关性上升降低高级层价值但提高股权层价值

14. How many of the following statements concerning the capital structure in a securitization are most likely correct? I. The mezzanine tranche is typically the smallest tranche size. II. The mezzanine and equity tranches typically offer fixed coupons. III. The senior tranche typically receives the lowest coupon.

- A. No statements are correct
- B. One statement is correct
- C. Two statements are correct
- D. Three statements are correct

答案:B

解析：

B 选项正确。这个说法正确：

- 陈述 I 错误：
 - 夹层级不是最小的层级
 - 股权层通常是最小的层级
- 陈述 II 错误：
 - 股权层没有固定票息
 - 股权层获得剩余现金流
- 陈述 III 正确：
 - 高级层被认为最安全
 - 因此获得最低票息

考察：证券化产品中高级层获得最低票息，股权层最小且无固定票息

15. A firm is experiencing financial difficulties. Using a contingent claims approach, which of the following best describes the valuation of their senior and subordinated debt?

- A. Both the senior debt and subordinated debt have positive exposures to debt maturity, firm volatility, and interest rates
- B. The senior debt has negative exposures to debt maturity, firm volatility, and interest rates. The subordinated debt has positive exposures to debt maturity, firm volatility, and interest rates
- C. The senior debt has positive exposures to debt maturity, firm volatility, and interest rates. The subordinated debt has negative exposures to debt maturity, firm volatility, and interest rates

D. Both the senior debt and subordinated debt have negative exposures to debt maturity, firm volatility, and interest rates

答案:B

解析:

A 选项错误。这个说法不正确:

- 高级债对这些因素有负向敞口
- 这个选项误解了高级债的特征
- 这个选项完全错误

B 选项正确。这个说法正确:

- 高级债特征:
 - 对到期期限有负向敞口
 - 对公司波动率有负向敞口
 - 对利率有负向敞口
- 次级债特征:
 - 在财务困境时表现类似股权
 - 对到期期限有正向敞口
 - 对公司波动率有正向敞口
 - 对利率有正向敞口

C 选项错误。这个说法不正确:

- 高级债对这些因素有负向敞口
- 次级债在财务困境时对这些因素有正向敞口
- 这个选项完全误解了债务特征

D 选项错误。这个说法不正确:

- 次级债在财务困境时对这些因素有正向敞口
- 这个选项误解了次级债的特征
- 这个选项部分正确但整体错误

考察: 财务困境时高级债有负向敞口, 次级债有正向敞口

16. Which of the following methods is typically implemented as the second stage of principal component analysis?

- A. Factor analysis
- B. Divisive clustering
- C. Hierarchical clustering
- D. The canonical correlation method

答案:A

解析:

A 选项正确。这个说法正确:

- 主成分分析的第二阶段:
 - 第一步是对主成分进行标准化

- 第二步是对新变量（因子载荷）进行标准化
- 因子分析是常用的第二阶段方法

B 选项错误。这个说法不正确：

- 分裂聚类不是主成分分析的第二阶段
- 这是另一种聚类方法
- 这个选项与主成分分析无关

C 选项错误。这个说法不正确：

- 层次聚类不是主成分分析的第二阶段
- 这是另一种聚类方法
- 这个选项与主成分分析无关

D 选项错误。这个说法不正确：

- 典型相关方法不是主成分分析的第二阶段
- 这是另一种统计分析方法
- 这个选项与主成分分析无关

考察：主成分分析的第二阶段通常使用因子分析方法

17. Under the contingent claim approach to the firm's capital structure, which of the following statements is true? Assume the amount of senior debt, subordinated debt, and equity is represented as F , U , and S , respectively.

- A. The value of subordinated debt is less than the value of senior debt
- B. Subordinated debt can be represented by a long call with exercise price of F and short call
- C. Subordinated debt behaves more like equity in distress and more like debt when the firm is not in distress
- D. The value of subordinated debt is always greater than the value of equity

答案:C

解析：

A 选项错误。这个说法不正确：

- 次级债价值不一定小于高级债
- 相对金额可能差异很大
- 这个选项过于绝对

B 选项错误。这个说法不正确：

- 次级债的期权表示错误
- 应该是执行价格为 F 的看涨期权多头
- 和执行价格为 $F+U$ 的看涨期权空头
- 这个选项误解了期权组合

C 选项正确。这个说法正确：

- 财务困境时：
 - 股权价值较小
 - 次级债表现类似股权

- 更接近获得剩余现金流
- 非财务困境时：
 - 公司价值较高
 - 次级债表现类似传统债务

D 选项错误。这个说法不正确：

- 次级债价值不一定大于股权
- 相对金额可能差异很大
- 这个选项过于绝对

考察：财务困境时次级债表现类似股权，非困境时表现类似债务

18. Harrison Michaels, FRM, an analyst at Hudson Risk Analytics, is discussing the default sensitivities of equity, mezzanine, and senior tranches and makes the following statements. Which of the following statements is(are) most likely correct? I. Default sensitivities are largest close to the attachment point between tranches. II. Default sensitivities are computed by shocking the credit default swap (CDS) default curve.

- A. I only
- B. II only
- C. Both I and II
- D. Neither I nor II

答案:A

解析：

A 选项正确。这个说法正确：

- 陈述 I 正确：
 - 违约敏感性在层级连接点附近最大
 - 这是因为损失接近连接点时影响最大
 - 所有层级的违约敏感性都是正值
 - 高违约率时敏感性趋近于零
- 陈述 II 错误：
 - 违约敏感性不是通过冲击 CDS 违约曲线计算
 - 而是通过冲击各种假设的违约概率计算
 - 这个陈述误解了计算方法

考察：违约敏感性在层级连接点附近最大，通过冲击违约概率计算

19. Advanced Pharmaceuticals is considering an investment in a very risky drug therapy treatment. If the investment is successful, Advanced Pharmaceuticals can earn substantial profits. On the other hand, if the therapy treatments create long-term side-effects, the firm will be subject to expensive litigation. How should Advanced Pharmaceuticals structure its investment to minimize its cost of capital?

- A. Ring-fence the investment so a special purpose entity (SPE) can issue debt at lower costs
- B. Ring-fence the investment into a SPE so the parent company can increase transparency
- C. Securitize the investment due to its predictable cash flows

D. Sell the investment in a true sale, increasing adverse selection

答案:B

解析:

A 选项错误。这个说法不正确:

- SPE 本身包含高风险投资
- 发行债务成本会更高
- 这个选项误解了 SPE 的作用

B 选项正确。这个说法正确:

- 通过 SPE 隔离风险投资:
 - 母公司剩余资产更透明
 - 可以以更优惠条件发行债务
 - 降低资本成本
- 真实销售可以:
 - 在法律上分离风险投资
 - 提高透明度
 - 减少逆向选择

C 选项错误。这个说法不正确:

- 该投资没有可预测的现金流
- 不像抵押贷款或汽车贷款
- 这个选项误解了证券化的条件

D 选项错误。这个说法不正确:

- 真实销售会减少逆向选择
- 不会增加逆向选择
- 这个选项误解了真实销售的影响

考察: 通过 SPE 隔离风险投资可提高透明度, 降低资本成本

20. Harris Smith, CFO of XYZ Bank Corp, is considering a \$500 million loan securitization. He has enlisted a well-respected structuring agent to help decide on the most beneficial structure. XYZ is a \$100 billion regional bank with a moderately strong balance sheet. Its current credit rating on unsecured debt is BBB. It recently issued a secured bond issue with a credit rating of A after ring-fencing certain assets. XYZ desires to minimize the cost of funds and achieve AAA credit rating on the senior tranche of the new securitization. After reviewing the financials of XYZ and forecasting future economic conditions, the structure has recommended an arbitrage CDO with the following loss distributions: Equity tranche: 0-30%. Junior tranche: 30-50%. Smith should use which of the following CDO structures?

- A. Arbitrage CDO with \$25 million equity tranche
- B. Arbitrage CDO with \$150 million equity tranche
- C. Balance sheet CDO with \$25 million equity tranche
- D. Balance sheet CDO with \$150 million equity tranche

答案:C

解析:

A 选项错误。这个说法不正确:

- 银行要证券化自己发放的贷款
- 应该使用资产负债表 CDO
- 这个选项选择了错误的 CDO 类型

B 选项错误。这个说法不正确：

- 银行要证券化自己发放的贷款
- 应该使用资产负债表 CDO
- 股权层级过大
- 这个选项完全错误

C 选项正确。这个说法正确：

- 资产负债表 CDO：
 - 适合证券化银行自身贷款
 - 符合银行的需求
- 股权层级：
 - \$25 百万占发行总额的 5%
 - 规模适中，可以保持高级层 AAA 评级
 - 有助于最小化融资成本

D 选项错误。这个说法不正确：

- 资产负债表 CDO 类型正确
- 但股权层级过大
- 可能影响高级层评级
- 这个选项部分正确但整体错误

考察：银行应使用资产负债表 CDO，股权层级占 5% 以保持高级层 AAA 评级

21. Which of the following statements regarding subprime mortgages is true? I. Senior tranches receive the highest return due to overcollateralization. II. Corporate bonds and subprime pools with the same credit rating will exhibit the same variation in losses.

- A. I only
- B. II only
- C. Both I and II
- D. Neither I nor II

答案:D

解析：

A 选项错误。这个说法不正确：

- 陈述 I 错误：
 - 高级层获得最低收益
 - 这是因为其在资本结构中的优先级
 - 超额抵押提供保护而非提高收益

- 陈述 II 错误：
 - 相同评级下损失波动性不同
 - ABS 结构损失波动性更大

B 选项错误。这个说法不正确：

- 陈述 I 错误
- 陈述 II 错误
- 这个选项完全错误

C 选项错误。这个说法不正确：

- 陈述 I 错误
- 陈述 II 错误
- 这个选项完全错误

D 选项正确。这个说法正确：

- 陈述 I 错误：
 - 高级层获得最低收益
 - 超额抵押提供保护而非提高收益
- 陈述 II 错误：
 - 相同评级下损失波动性不同
 - ABS 结构损失波动性更大

考察：次贷证券化中高级层收益最低，ABS 结构损失波动性更大

22. The manager makes additional observations as follows: • The loans in the collateral pool pay a fixed spread of 2.2% • There were no defaults in year 1 • The value of the overcollateralization account at the end of year 1 was EUR 0 What is the value of the overcollateralization account at the end of year 2 if there are 8 defaults in year 2?

- A. EUR 600,000
- B. EUR 1,056,000
- C. EUR 2,544,000
- D. EUR 3,600,000

答案:C

解析：

A 选项错误。这个说法不正确：

- 计算错误
- 低估了超额抵押账户的价值
- 这个选项完全错误

B 选项错误。这个说法不正确：

- 计算错误
- 低估了超额抵押账户的价值
- 这个选项完全错误

C 选项正确。这个说法正确：

- 计算过程：
 - 贷款利差为 2.2
 - 第一年无违约
 - 第二年 8 笔违约
 - 超额抵押账户价值 = EUR 2,544,000

D 选项错误。这个说法不正确：

- 计算错误
- 高估了超额抵押账户的价值
- 这个选项完全错误

考察：超额抵押账户价值取决于贷款利差和违约情况

4 考点 4: 证券化

1. Assume an MBS is composed of the following four different pools of mortgages: \$2 million of mortgages that have a maturity of 90 days. \$3 million of mortgages that have a maturity of 180 days. \$5 million of mortgages that have a maturity of 270 days, \$10 million of mortgages that have a maturity of 360 days. What is the weighted average maturity (WAM) of these mortgage pools?

- A. 167 days
- B. 225 days
- C. 252 days
- D. 284 days

答案:D

解析:

A 选项错误。这个说法不正确:

- 计算错误
- 低估了加权平均期限
- 这个选项完全错误

B 选项错误。这个说法不正确:

- 计算错误
- 低估了加权平均期限
- 这个选项完全错误

C 选项错误。这个说法不正确:

- 计算错误
- 低估了加权平均期限
- 这个选项完全错误

D 选项正确。这个说法正确:

- 计算过程:
 - 总金额 = \$2M + \$3M + \$5M + \$10M = \$20M
 - $WAM = (90 \times 2 + 180 \times 3 + 270 \times 5 + 360 \times 10) / 20$
 - $= (180 + 540 + 1350 + 3600) / 20$
 - = 284 天

考察: 抵押贷款支持证券的加权平均期限

2. Suppose the annual prepayment rate CPR for a mortgage-backed security is 6 percent. What is the corresponding single-monthly mortality rate SMM?

- A. 0.514%
- B. 0.334%
- C. 0.500%
- D. 1.355%

答案:A

解析:

A 选项正确。这个说法正确:

- 计算过程: $CPR = 6\%$ $SMM = 1 - (1 - CPR)^{1/12} = 1 - (1 - 0.06)^{1/12} = 0.514\%$

考察: 年提前还款率 6% 对应的月提前还款率为 0.514%

3. Securitized products are often customized to meet the needs of the investor as well as the originator. What type of asset-backed securities (ABSs) typically uses a revolving structure?

- A. Residential mortgage
- B. Credit card debt
- C. Commercial mortgage
- D. Commercial paper

答案:B

解析:

A 选项错误。这个说法不正确:

- 住宅抵押贷款不使用循环结构
- 有固定的还款计划
- 这个选项与循环结构无关

B 选项正确。这个说法正确:

- 信用卡债务使用循环结构:
 - 没有预先设定的还款计划
 - 本金以大额方式偿还
 - 分为循环期和摊销期
 - 循环期将本金用于购买新贷款
 - 摊销期进行最终偿付

C 选项错误。这个说法不正确:

- 商业抵押贷款不使用循环结构
- 有固定的还款计划
- 这个选项与循环结构无关

D 选项错误。这个说法不正确:

- 商业票据不使用循环结构
- 有固定的到期日
- 这个选项与循环结构无关

考察: 信用卡债务证券化通常使用循环结构, 分为循环期和摊销期

4. Which of the following statements regarding credit enhancements in the process of structuring a securitization through a special purpose vehicle (SPV) is correct?

- A. The securitization process is structured such that the asset side of the SPV has a lower cost than the liability side of the SPV

- B. Credit enhancements are typically only associated with mortgage-backed securities (MBS) and are not used in other types of asset-backed securities (ABS)
- C. The most senior class of notes is often overcollateralized in order to reduce the risk of the asset-backed security (ABS)
- D. A margin step-up is sometimes used by an asset-backed securities (ABS) where the coupon structure increases after a call date

答案:D

解析:

A 选项错误。这个说法不正确:

- SPV 的负债成本低于资产成本
- 这样才能产生超额利差
- 这个选项与实际情况相反

B 选项错误。这个说法不正确:

- 信用增级不仅用于 MBS
- 也用于其他类型的 ABS
- 这个选项过于局限

C 选项错误。这个说法不正确:

- 最低层级票据通常被超额担保
- 不是最高层级
- 这个选项误解了超额担保的对象

D 选项正确。这个说法正确:

- ABS 可能使用利差递增:
 - 在赎回日期后提高票息结构
 - 这是一种信用增级方式
 - 有助于提高证券的吸引力

考察: ABS 可能使用利差递增作为信用增级方式

5. A major benefit of securitization for a financial institution is the ability to remove assets from the balance sheet, which lowers risk and the required regulatory capital. While a large portion of the risk is removed from the balance sheet the originating financial institution often maintains a portion of the risk. Which of the following terms best identify the risk that is maintained by the originator?

- A. Correlation
- B. Excess spread
- C. First-loss piece
- D. Guarantor of collateral value

答案:C

解析:

A 选项错误。这个说法不正确:

- 相关性不是发起人保留的风险

- 这是资产池的特征
- 这个选项与风险保留无关

B 选项错误。这个说法不正确：

- 超额利差不是发起人保留的风险
- 这是信用增级方式
- 这个选项与风险保留无关

C 选项正确。这个说法正确：

- 第一损失部分：
 - 发起人通常保留所有权
 - 信用质量最低的资产类别
 - 违约时最先承担损失
 - 代表发起人保留的风险

D 选项错误。这个说法不正确：

- 抵押品价值担保不是发起人保留的风险
- 这是外部信用增级方式
- 这个选项与风险保留无关

考察：发起人通常保留第一损失部分的所有权

6. Which of the following is an example of internal enhancements?

- A. CDS
- B. Put options on assets
- C. Overcollateralization
- D. Letters of credit

答案:C

解析：

A 选项错误。这个说法不正确：

- CDS 是外部信用增级
- 需要第三方参与
- 这个选项与内部增级无关

B 选项错误。这个说法不正确：

- 资产看跌期权是外部信用增级
- 需要第三方参与
- 这个选项与内部增级无关

C 选项正确。这个说法正确：

- 内部增级方式包括：
 - 超额担保
 - 直接股权发行

- 自留
- 现金担保账户
- 超额利差

D 选项错误。这个说法不正确：

- 信用证是外部信用增级
- 需要银行参与
- 这个选项与内部增级无关

考察：超额担保是证券化产品的内部增级方式

7. Under a 200% PSA model, what is the single monthly mortality (SMM) for, respectively, month 10 and month 30?

- A. 0.12% and 0.36%
- B. 0.17% and 0.51%
- C. 0.28% and 0.65%
- D. 0.34% and 1.06%

答案:D

解析：

D 选项正确。这个说法正确：

- 计算过程：
 - 基础 PSA 模型：每月增加 0.2% CPR，直到第 30 个月达到 6%
 - 第 10 个月：100% PSA = $0.2\% \times 10 = 2.0\%$
 - 第 10 个月：200% PSA = 4.0%
 - $SMM = 1 - (1 - 4.0\%)^{(1/12)} = 0.34\%$
 - 第 30 个月：100% PSA = $0.2\% \times 30 = 6.0\%$
 - 第 30 个月：200% PSA = 12.0%
 - $SMM = 1 - (1 - 12.0\%)^{(1/12)} = 1.06\%$

考察：PSA 模型下的月提前还款率

8. The maximum benefits to the buyer of a credit-linked note (CLN) accrue when:

- A. There is a small credit downgrade
- B. There is no credit downgrade
- C. There is a large credit downgrade
- D. There is a default

答案:B

解析：

B 选项正确。这个说法正确：

- CLN 购买者的最大收益：
 - 没有信用降级或违约时
 - 获得高额回报

- 主要风险是降级或违约
- 违约可能导致血本无归

考察：信用联结票据在没有信用降级时获得最大收益

9. Which of the following participants in the securitization process is least likely to face a conflict of interest?

- A. Credit rating agency and servicer
- B. Servicer and underwriter
- C. Custodian and trustee
- D. Trustee and manager

答案:C

解析：

A 选项错误。这个说法不正确：

- 评级机构和服务商存在利益冲突
- 评级机构需要客观评级
- 服务商可能影响评级结果

B 选项错误。这个说法不正确：

- 服务商和承销商存在利益冲突
- 服务商负责资产服务
- 承销商负责证券销售

C 选项正确。这个说法正确：

- 托管人和受托人：
 - 在证券化过程中作用最小
 - 主要负责资产保管和监督
 - 不参与核心业务决策
 - 利益冲突最小

D 选项错误。这个说法不正确：

- 受托人和管理人存在利益冲突
- 受托人负责监督
- 管理人负责投资决策

考察：托管人和受托人在证券化过程中利益冲突最小

10. Which of the following measures are most likely to be used by a securitized product backed by student loans?

- A. Single monthly mortality (SMM), constant prepayment rate (CPR), and Public Securities Association (PSA)
- B. Loss curves and absolute prepayment speed (APS)
- C. Weighted average life (WAL), weighted average maturity (WAM), and weighted average coupon (WAC)
- D. Debt service coverage ratio (DSCR) and monthly payment rate (MPR)

答案:A

解析:

A 选项正确。这个说法正确:

- 学生贷款证券化常用指标:
 - 月提前还款率 (SMM)
 - 固定提前还款率 (CPR)
 - PSA 提前还款模型
 - 这些指标适用于学生贷款和抵押贷款

B 选项错误。这个说法不正确:

- 损失曲线和绝对提前还款速度
- 不适用于学生贷款证券化
- 这个选项与题目无关

C 选项错误。这个说法不正确:

- 加权平均期限和加权平均票息
- 是资产池的基本特征
- 不是提前还款指标

D 选项错误。这个说法不正确:

- 偿债覆盖率 and 月还款率
- 不适用于学生贷款证券化
- 这个选项与题目无关

考察: 学生贷款证券化使用 SMM、CPR 和 PSA 指标

11. As the CRO of a retail bank, you are presenting to your Risk Committee the benefits of securitizing a pool of mortgages. Which of the following would you use to support your arguments that this will benefit the bank?

1. It will improve your bank’s return on capital
2. Will immediately increase your bank’s available capital
3. You will be able to offer an attractive yield to investors
4. It will lower your borrowing costs

- A. I and III only
- B. I, III, and IV only
- C. I, II, III and IV
- D. I, II, and IV only

答案:C

解析:

A 选项错误。这个说法不正确:

- 只包含了部分好处
- 忽略了资本释放和融资成本降低

- 这个选项不完整

B 选项错误。这个说法不正确：

- 只包含了部分好处
- 忽略了资本释放
- 这个选项不完整

C 选项正确。这个说法正确：

- 证券化的所有好处：
 - 提高资本回报率
 - 立即增加可用资本
 - 为投资者提供有吸引力的收益
 - 降低融资成本
 - 优化资产负债表结构
 - 提高资产池透明度

D 选项错误。这个说法不正确：

- 只包含了部分好处
- 忽略了投资者收益
- 这个选项不完整

考察：抵押贷款证券化可以带来资本回报、资本释放、投资者收益和成本降低

12. Which of the following statements was not one of the flaws in the securitization process prior to the start of the credit crisis in 2007?

- A. An active originate-to-distribute model where a strong profit motive took precedence over ethical lending and underwriting
- B. The securitized products were so opaque that investors could not evaluate the true risks of the investment
- C. Structured investment vehicles (SIVs) were used to enhance the risk discovery process for investors and regulators
- D. Banks held securitized assets in off-balance-sheet entities, thus further masking the true risks in the system

答案:C

解析：

A 选项错误。这个说法不正确：

- 这是 2007 年危机前的缺陷之一
- 利润动机优先于道德放贷
- 这个选项是真实缺陷

B 选项错误。这个说法不正确：

- 这是 2007 年危机前的缺陷之一
- 产品不透明导致风险难以评估
- 这个选项是真实缺陷

C 选项正确。这个说法正确：

- SIVs 实际上：

- 增加了不透明度
- 是银行的表外实体
- 使投资者难以审查
- 不是用于增强风险发现

D 选项错误。这个说法不正确：

- 这是 2007 年危机前的缺陷之一
- 表外实体掩盖了真实风险
- 这个选项是真实缺陷

考察：SIVs 不是用于增强风险发现，而是增加了不透明度

13. In a securitized transaction, over-collateralization results when:

- A. The originator puts aside some cash in a reserve account to absorb credit losses
- B. A securitization transaction carves up the cash flows generated from the asset pool into various pieces
- C. The interest payments and other fees received on the assets in the pool exceed the interest payment made on the asset-backed security (ABS) plus the fee paid to service the assets along with miscellaneous expenses
- D. The value of the assets in the pool exceeds the amount of asset-backed security(ABS) involved

答案:D

解析：

A 选项错误。这个说法不正确：

- 这是现金储备账户的定义
- 用于吸收信用损失
- 不是超额抵押

B 选项错误。这个说法不正确：

- 这是分级结构的定义
- 将现金流分成不同层级
- 不是超额抵押

C 选项错误。这个说法不正确：

- 这是超额利差的定义
- 资产收益超过证券支出
- 不是超额抵押

D 选项正确。这个说法正确：

- 超额抵押的定义：
 - 资产池价值超过证券发行规模
 - 实际发行的证券规模小于资产规模
 - 是一种信用增级方式

考察: 超额抵押是指资产池价值超过证券发行规模

14. A pass-through mortgage-backed security (MBS) with a weighted average maturity (WAM) of 30 months has an original principle balance of \$2.0 billion. If the valuation model assumes a 300% PSA prepayment speed, which is nearest to the first month’s prepaid (not scheduled) principal?

- A. \$519,000
- B. \$833,000
- C. \$1.00 million
- D. \$2.15 million

答案:C

解析:

C 选项正确。这个说法正确:

- 计算过程:
 - $300\% \text{ PSA} = 3 \times \text{CPR} = 3 \times 0.2\% = 0.6\%$
 - $\text{SMM} = 1 - (1 - \text{CPR})^{(1/12)} = 1 - (1 - 0.006)^{(1/12)} = 0.00050138$
 - $\text{提前还款金额} = 0.00050138 \times 2,000,000,000 = 1,002,760$
 - 最接近 100 万美元

考察: 300%PSA 下首月提前还款金额约为 100 万美元

15. A risk consultant is reviewing a proposal for a US-based startup credit card company. A member of the senior management asks about the benefit of securitizing the company’s loan book and the type of asset securitization structure that would best suit the credit card debt portfolio. If the consultant recommends securitization, which securitization structure would be most appropriate and why?

- A. An amortizing structure, because principal payments from the credit assets are paid to investors in a series of equal installments or in a lumpsum at maturities
- B. A master trust structure, because the structure allows multiple securitizations to be issued, and notes are issued out of the credit asset pool based on investor demand
- C. An amortizing structure, because the credit assets have relatively long maturities and low prepayment speeds
- D. A master trust structure, because the structure assumes no change in prepaymert speed, and investors are paid principal and interest on a coupon-by-coupon basis

答案:B

解析:

A 选项错误。这个说法不正确:

- 信用卡债务不适合分期偿还结构
- 信用卡没有固定还款计划
- 这个选项与信用卡证券化无关

B 选项正确。这个说法正确:

- 主信托结构的优势:
 - 可以发行多个证券化产品
 - 根据投资者需求发行票据

- 适合信用卡债务的特点
- 灵活性更高

C 选项错误。这个说法不正确：

- 信用卡债务期限不固定
- 提前还款速度不低
- 这个选项与信用卡证券化无关

D 选项错误。这个说法不正确：

- 主信托结构不假设固定提前还款速度
- 不是按票息支付本息
- 这个选项描述不准确

考察：信用卡债务证券化最适合使用主信托结构

16. A fixed-income portfolio manager at a financial institution is estimating the weighted average coupon (WAC) of an MBS pool before making an investment. The pool consists of four residential mortgage loans. Information about the loans at the beginning of a specified month is given below:

Loan	Outstanding beginning balance	Interest(coupon) rate	Scheduled principal payment	Remaining term (months)
1	USD 240,000	6.5%	USD 18,800	16
2	USD 525,000	5.0%	USD 23,000	34
3	USD 470,000	6.8%	USD 28,700	14
4	USD 875,000	3.0%	USD 23,600	60

What is the WAC of the MBS pool at the beginning of the month and what is the pool's ability to pay if the net coupon payable to investors in the MBS tranches is 4.76%?

- A. WAC is 4.44% and the pool's ability to pay is relatively weak
- B. WAC is 4.74% and the pool's ability to pay is relatively weak
- C. WAC is 4.77% and the pool's ability to pay is relatively strong
- D. WAC is 5.35% and the pool's ability to pay is relatively strong

答案:B

解析：

A 选项错误。这个说法不正确：

- WAC 计算错误
- 低估了加权平均票息
- 这个选项完全错误

B 选项正确。这个说法正确：

- 计算过程：
 - 总余额 = 240,000 + 525,000 + 470,000 + 875,000 = 2,110,000
 - $WAC = (240,000 \times 6.5\% + 525,000 \times 5.0\% + 470,000 \times 6.8\% + 875,000 \times 3.0\%) / 2,110,000$
 - = 4.74%
 - 支付能力较弱，因为 WAC(4.74%) < 投资者票息 (4.76%)

C 选项错误。这个说法不正确：

- WAC 计算错误
- 高估了支付能力
- 这个选项完全错误

D 选项错误。这个说法不正确：

- WAC 计算错误
- 高估了加权平均票息
- 这个选项完全错误

考察：加权平均票息低于投资者票息时，资产池支付能力较弱

17. A risk analyst at a hedge fund is assessing the following positions held by the fund:

- Contingent convertible bonds (CoCos) issued by GTU Bank; the triggering event for the conversion is when the price of GTU Bank’s publicly traded stock reaches a specified threshold
- Senior tranche of an MBS issued by a bankruptcy-remote special purpose vehicle; the mortgages in the securitization are originated by GTU Bank
- Interest rate swap contract as the fixed-rate payer, with Bank NPV as the floating-rate payer; Bank NPV posts high-quality government bonds as collateral

The analyst considers a scenario under which interest rates increase and the stock price of GTU Bank decreases. As the analyst reviews the impact of these changes in market conditions on the three positions, which of the following is correct?

- A. The CoCos held by the hedge fund are less likely to be converted to stocks of GTU Bank
- B. The value of the senior tranche of the MBS held by the hedge fund will increase
- C. The hedge fund’s wrong-way risk in the interest rate swap will decrease
- D. The value of the interest rate swap from the perspective of the hedge fund will increase

答案:D

解析：

A 选项错误。这个说法不正确：

- 股价下跌会增加转换可能性
- 触发条件更容易满足
- 这个选项与实际情况相反

B 选项错误。这个说法不正确：

- 利率上升会降低 MBS 价值
- 提前还款风险增加
- 这个选项与实际情况相反

C 选项错误。这个说法不正确：

- 利率上升会增加错向风险
- 对手方信用风险增加
- 这个选项与实际情况相反

D 选项正确。这个说法正确：

- 利率互换价值变化：
 - 作为固定利率支付方
 - 利率上升时价值增加
 - 对手方提供高质量抵押品
 - 信用风险较小

考察：利率上升时固定利率支付方的互换价值增加